



11 Physics

Wave Properties

THE NATURE OF WAVES

Waves have the following characteristics.

1. They consist of disturbances progressing through all or part of a medium.
2. An energy change or source of energy is needed to set up the waves.
3. **Energy is transferred** as the wave progresses through the medium.
4. The medium itself is not moved from one point to another.

Example

On the ocean, a boat bobs up and down as the waves move underneath it. The boat does not move with the wave - energy is being transmitted but the medium does not move with it.

The water particles merely oscillate up and down about some equilibrium position in accordance to the energy as it reaches them.

Mechanical waves (e.g. water waves, sound) need a medium to travel in, while **electromagnetic waves** (e.g. light) do not need a medium.

MECHANICAL WAVES

This type of wave needs an **energy source to produce a disturbance** and an **elastic medium to transmit the disturbance**.

An **elastic medium** has particles that occupy a mean position from which they can be temporarily displaced by an outside force. They **return to their original position** after the force has ceased.

e.g. Air particles behave elastically.

There are two types of mechanical waves - **longitudinal** and **transverse**.

Longitudinal Wave (e.g. sound)

- The particles of the medium move **parallel** to the direction of propagation of the wave disturbance.
- Longitudinal waves are a series of **compressions** and **rarefactions**.
- **Compression** - particles have been pushed closer together.
i.e. **High pressure area**.
- **Rarefaction** - particles are further apart.
i.e. **Low pressure area**.

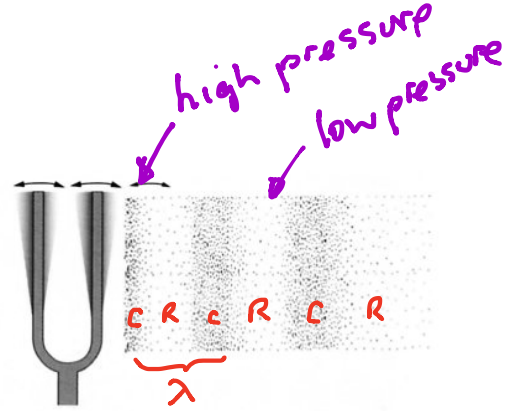
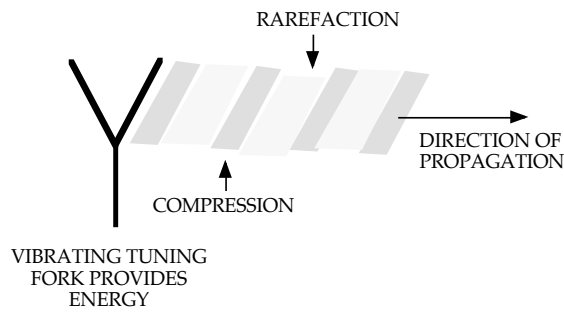


Figure 5.3

The rapid vibration of a tuning fork makes the air molecules around it vibrate in much the same way as the human larynx or a loudspeaker—creating the compressions and rarefactions of a pressure wave that we perceive as sound.

After the disturbance has passed a given point, the particles return to the equilibrium or mean position.

The distance a particle is displaced from its equilibrium position depends on the energy - the greater the energy, the greater the displacement.

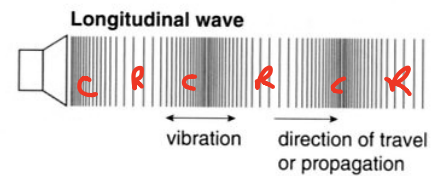
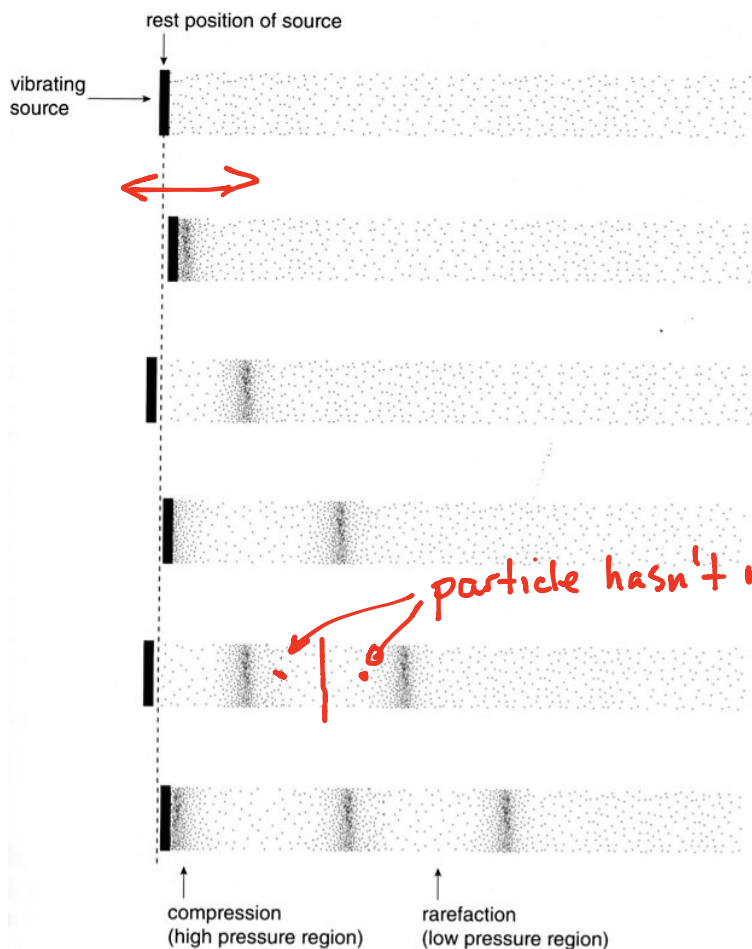


Figure 5.7

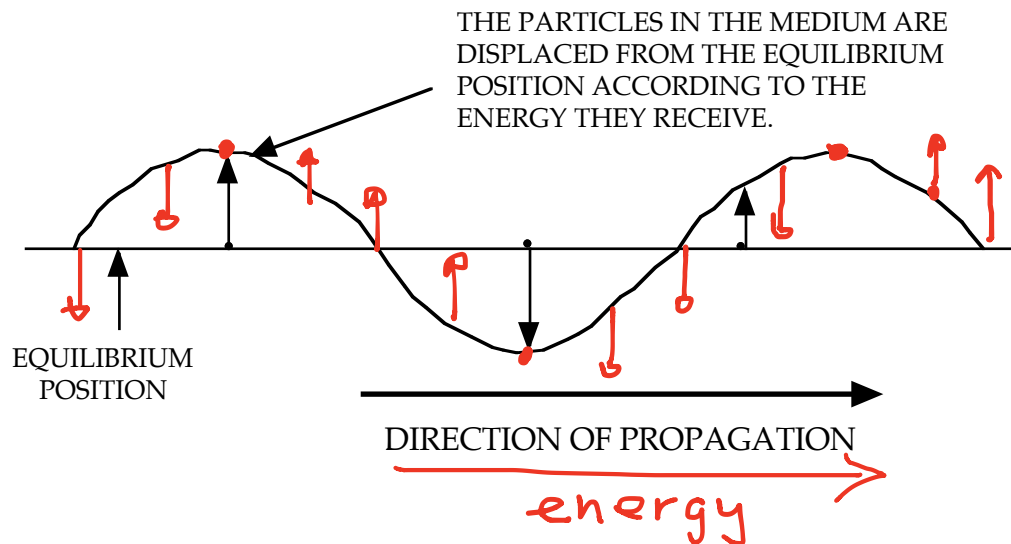
In a longitudinal wave, the vibrations of the individual particles are in the same direction as the direction of wave propagation.

Figure 5.8

In a medium in which there is no sound, the particles are evenly spread. As the source moves backwards and forwards, the continual vibration produces a series of compressions and rarefactions moving continuously away from the source. In this way, sound energy is propagated through a medium. Once the source ceases to vibrate, the particles in the medium return to approximately the same position. There is no net movement of the particles.

Transverse Waves (e.g. water waves)

- The particles of the medium move **perpendicular** to the direction of propagation of the wave disturbance.



- The particles vibrate about their equilibrium position in a periodic fashion.
i.e. They move up and down only - they don't move with the wave front as it moves through the medium.

In both transverse and longitudinal waves, **kinetic energy is transferred through the medium without the medium as a whole being displaced.**

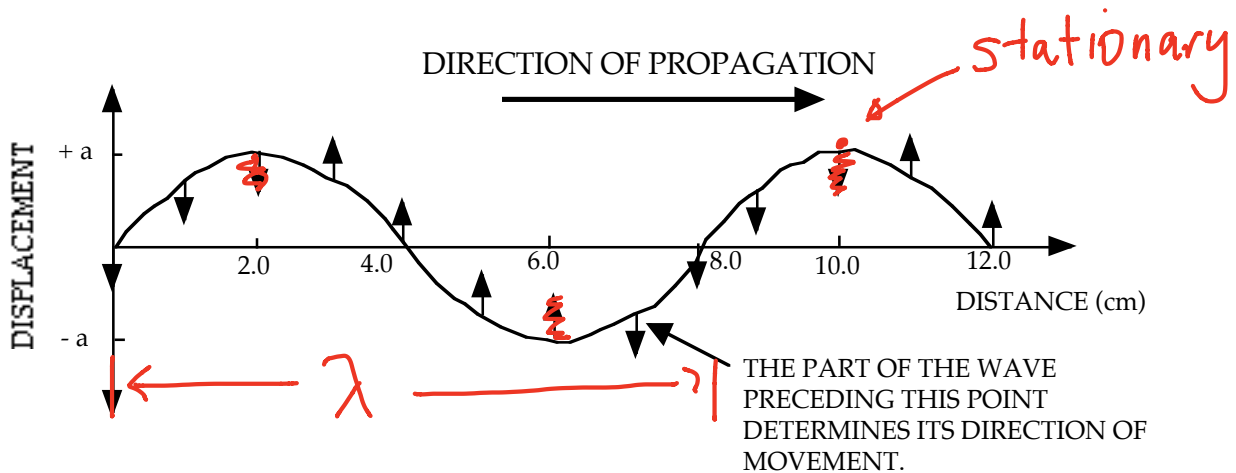
GRAPHING WAVE MOTION

There are two types of graphs that are drawn to represent a wave motion, namely *displacement - distance* and *displacement - time* graphs.

Displacement - Distance Graph

"photograph"

- This graph shows all of the particles of the medium at a given instant and their respective displacement from the equilibrium position.



- The arrows on the diagram above show the direction each particle is moving at that instant.
- The energy of the wave that is about to reach the point (i.e. the preceding part of the graph) determines whether the movement will be up or down.
- Even though longitudinal waves cause the particles of the medium to move backwards and forwards rather than up and down, this motion is still represented on a graph like the one above.

Displacement - Time Graph

- This graph represents one point in the medium and shows how it is moving with respect to time.



- This graph is used to represent both transverse and longitudinal wave movement.

$$f = \frac{1}{T}$$

CHARACTERISTICS OF WAVES

Phase

This is the *position and motion of a particle at any instant*.

Particles having the *same displacement from their equilibrium position* and *moving in the same direction* are in phase.

Wavelength (λ)

This is the distance between two successive particles that are in phase.

← "lambda"

Amplitude (a)

The maximum displacement of the particles from their equilibrium position.

Wave Speed (c)

The distance travelled by the wave per unit time.
i.e. It's a velocity.

← speed of light - can use V .

Frequency (f, ν)

The number of waves that pass a given point per unit time.

This is the same as the frequency of vibration of the energy source.

← said "gnu".

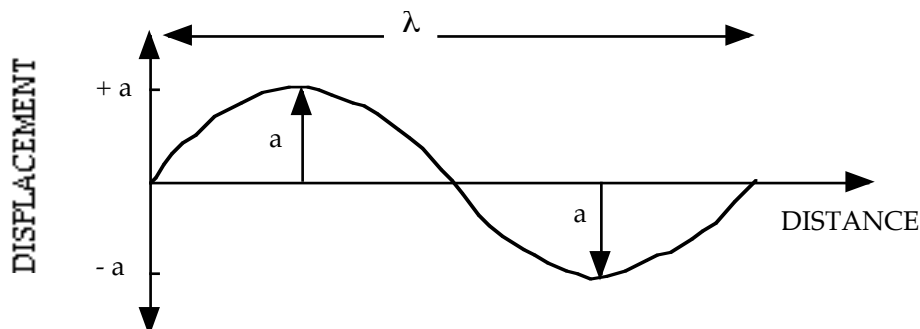
Period (T)

The time taken for one wavelength to pass a given point.

The time taken for a particle to undergo one complete oscillation.

$$T = 1/f \quad \text{or} \quad T = 1/\nu$$

where f, ν = frequency (Hz)
 T = period (s)



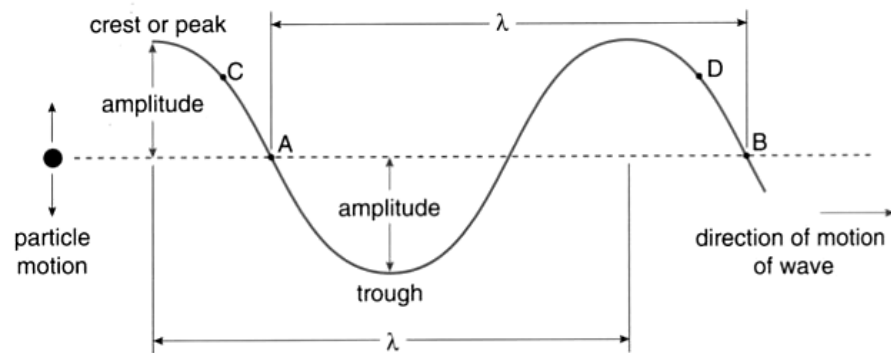
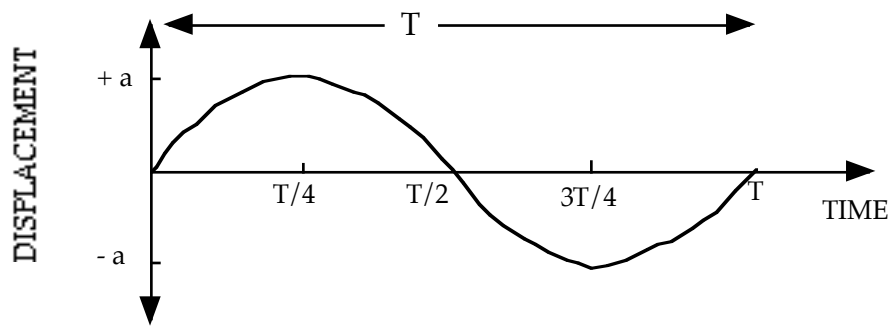


Figure 5.11

A simple pictorial representation of a transverse wave.

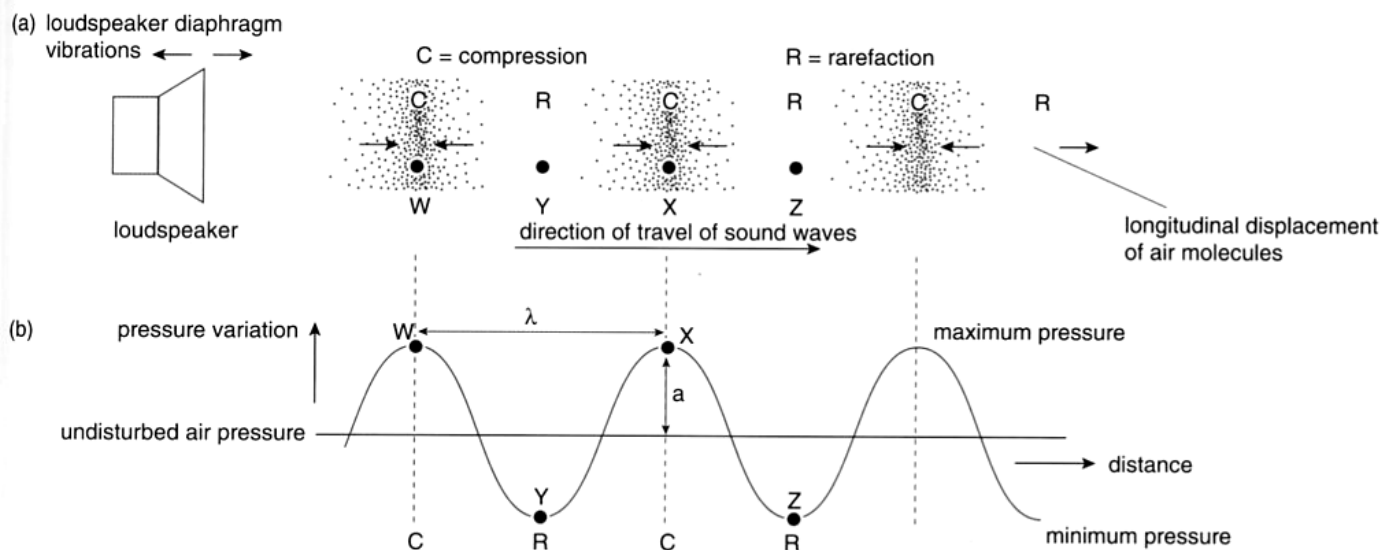


Figure 5.12

A longitudinal wave as a simple pictorial representation.

WAVE EQUATION

The velocity of a wave is given by:

$$c = f\lambda \quad \text{or} \quad c = v\lambda$$

$$v = f\lambda$$

where c = velocity of the wave (ms^{-1})
 f, v = frequency of the wave (Hz)
 λ = wavelength of the wave (m)

This equation applies to both mechanical and electromagnetic waves.

Examples

1. A stone is dropped into a still pool of water. It generates 20 waves that cover a distance of 10.0 m of the surface. The outer wave covers the 10.0 m in a time of 5.00 s and the average height of the waves is 10.0 mm (crest to trough).

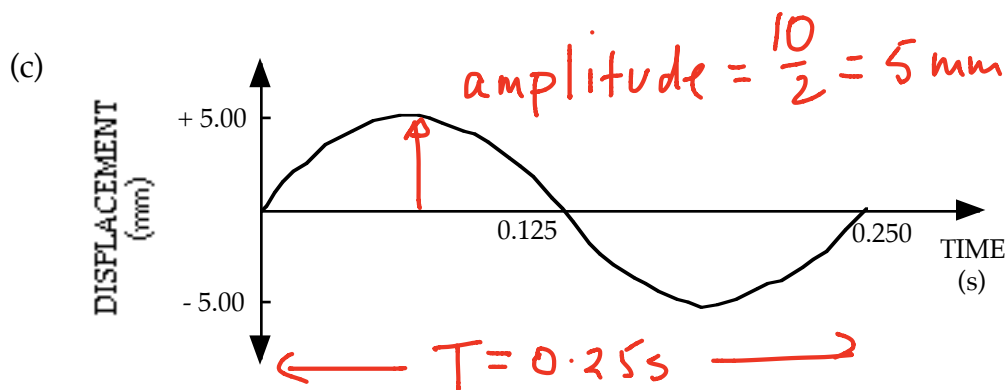
- (a) Determine the wavelength and velocity of the waves.
- (b) Calculate the period of the waves.
- (c) Draw a displacement - time graph to represent the motion of the waves.

$$\begin{aligned} \text{(a)} \quad \lambda &= \frac{10.0}{20} & v &= \frac{10.0}{5.00} & v &= \frac{s}{t} \\ &= 0.500 \text{ m} & &= 2.00 \text{ ms}^{-1} \end{aligned}$$

$\therefore$ Wavelength = 0.500 m and wave velocity = 2.00 ms⁻¹.

$$\begin{aligned} \text{(b)} \quad \Rightarrow \quad c &= f\lambda & T &= \frac{1}{f} \\ f &= \frac{v}{\lambda} & &= \frac{1}{4.00} \\ &= \frac{2.00}{0.500} & &= 0.250 \text{ s} \\ &= 4.00 \text{ Hz} \end{aligned}$$

$\therefore$ Period of the waves = 0.250 s .



2. Radio station JJJ operates at 99.3 MHz. What is the wavelength of the transmission?

$$\begin{aligned}
 c &= f\lambda \\
 \Rightarrow \lambda &= \frac{c}{f} \\
 &= \frac{3.00 \times 10^8}{99.3 \times 10^6} \\
 &= 3.021 \text{ m}
 \end{aligned}$$

RADIO WAVES ARE PART OF THE LIGHT SPECTRUM

$$\therefore \text{Wavelength} = 3.02 \text{ m}$$

3.

Waves are created on the surface of the water in a wave pool by a device that dips up and down 20.0 times per minute. The velocity of the resultant wave is 3.00 m s^{-1} .

a State the frequency, period and wavelength of the observed wave.

b What happens to the value of the velocity and wavelength if the frequency of the source is doubled?

Solution

a Since frequency is defined as the number of waves or cycles that pass a given point per second:

$$f = 20.0 \text{ per minute}$$

$$f = \frac{20.0}{60} = 0.333 \text{ Hz}$$

$$f = 0.333 \text{ Hz}$$

$$\begin{aligned}
 T &= \frac{1}{f} = \frac{1}{0.333} \\
 &= 3.00 \text{ s}
 \end{aligned}$$

The period is therefore 3.00 s

The wavelength is then:

$$f = 0.333 \text{ Hz}$$

$$v = f\lambda$$

$$v = 3.00 \text{ m s}^{-1}$$

$$\begin{aligned}
 \lambda &= \frac{v}{f} = \frac{3.00}{0.333} \\
 &= 9.00 \text{ m}
 \end{aligned}$$

b The velocity is determined by the properties of the medium and so it is unchanged, $v = 3.00 \text{ m s}^{-1}$. At a set velocity, doubling the frequency will result in wavelengths of half the original value, i.e. 4.50 m.

REFLECTION

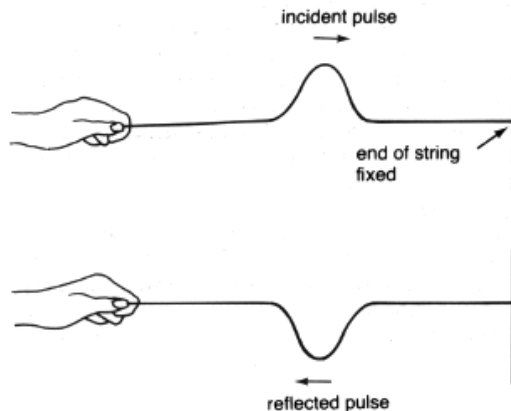
Waves are reflected by barriers, the amount of energy reflected being determined by the energy-absorbing nature of the barrier.

Reflecting surfaces can be classified as follows.

Rigid

Most energy reflected

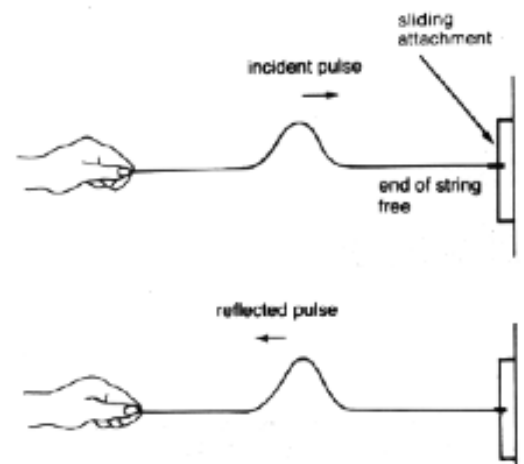
- Absorbs little or no energy - all is reflected.
e.g. Walls of a pool are rigid to any water waves.
String fixed at one end.
- The reflected wave is 180° out of phase to the incident wave.



The fixed end of the string represents a rigid surface (no energy is transmitted to the wall).

Non-Rigid

- Some energy is absorbed, the rest is reflected.
- The energy of the reflected wave is *less* than the energy of the incident wave.
- The reflected wave is *in phase* with the incident wave.
e.g. Carpet absorbs much of the incident sound energy, resulting in a lower level of reflected sound in a room.
- A string attached at a “free end” can represent a non-rigid surface.

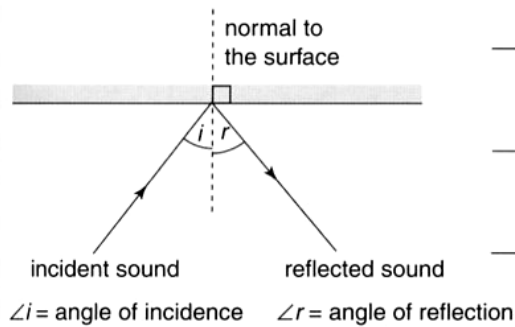


Laws Of Reflection

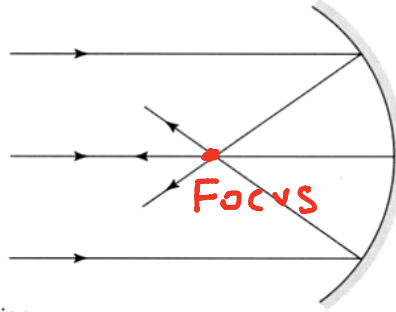
YR 9!

1. The angle of incidence (i) = The angle of reflection (r)
2. The incident ray and reflected ray are on opposite sides of the normal, and all three are in the same plane.

(a) Reflection at a plane barrier



(b) Reflection at a concave surface



(c) Reflection at a convex surface

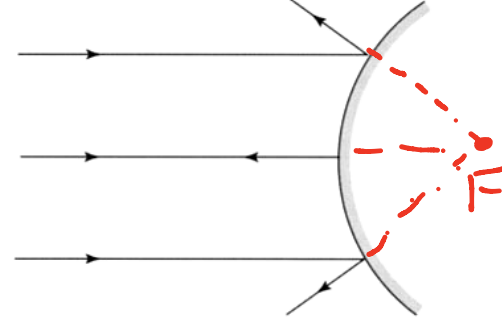
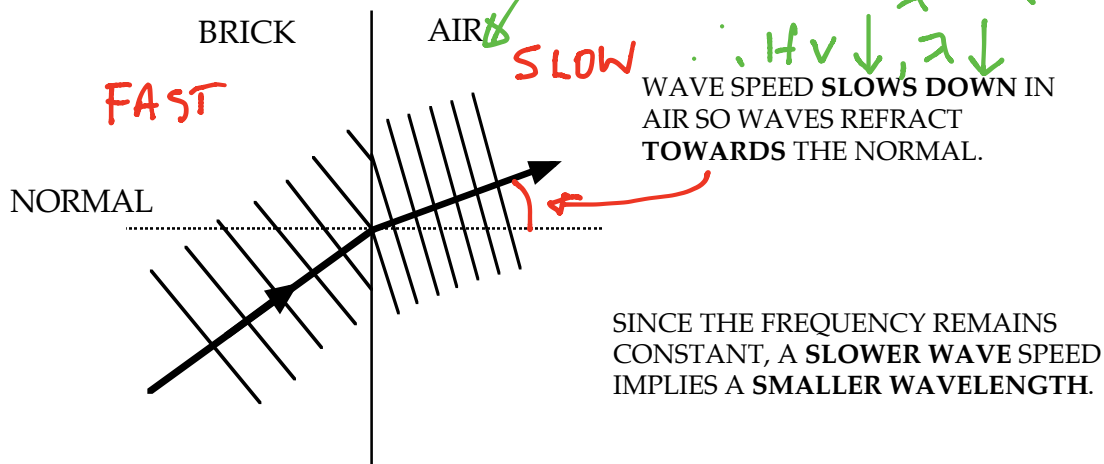


Figure 5.15

Ray tracings of the paths of reflected sound waves.

REFRACTION

Refraction occurs when there is a change in the velocity of a wave as it travels *obliquely* from one medium into another.

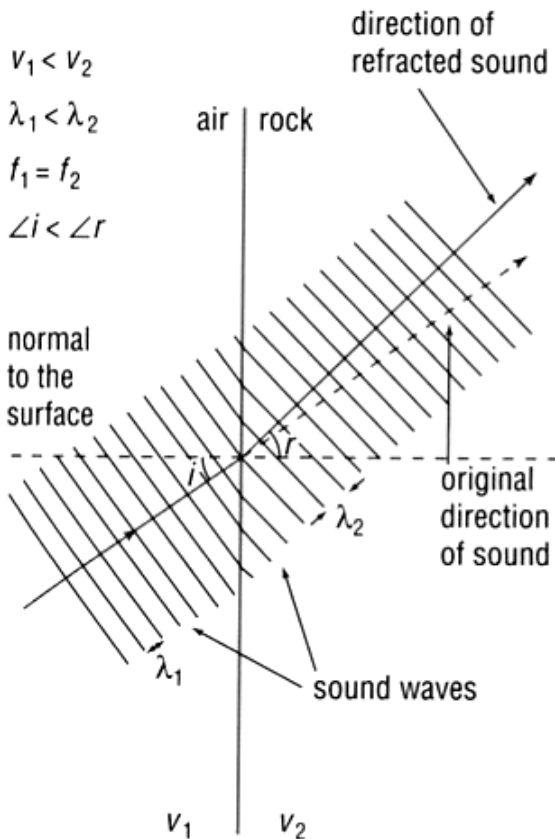


Slower speed in second medium $\Rightarrow$ smaller λ , *refract towards normal*.

Faster speed in second medium $\Rightarrow$ larger λ , *refract away from normal*.

If the incoming wave is *perpendicular to the boundary of the two media*, no refraction occurs.

(a) Sound being refracted away from the normal



(b) Sound being refracted towards the normal

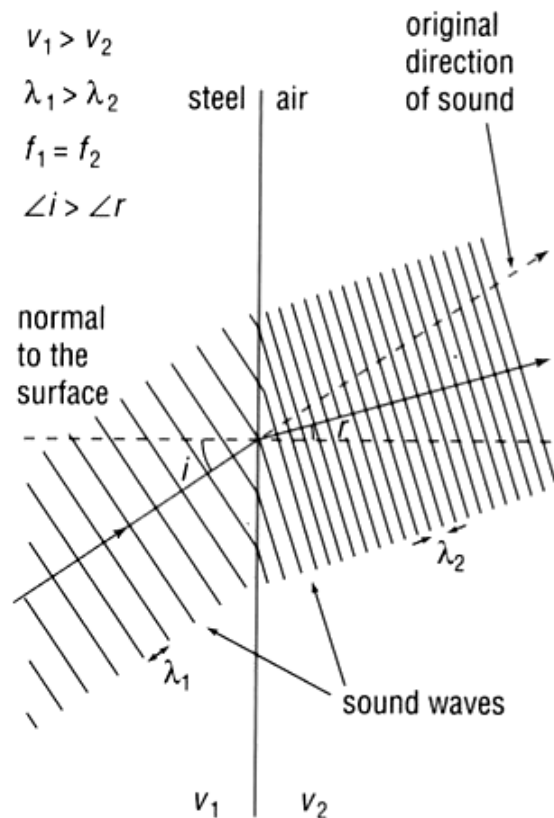
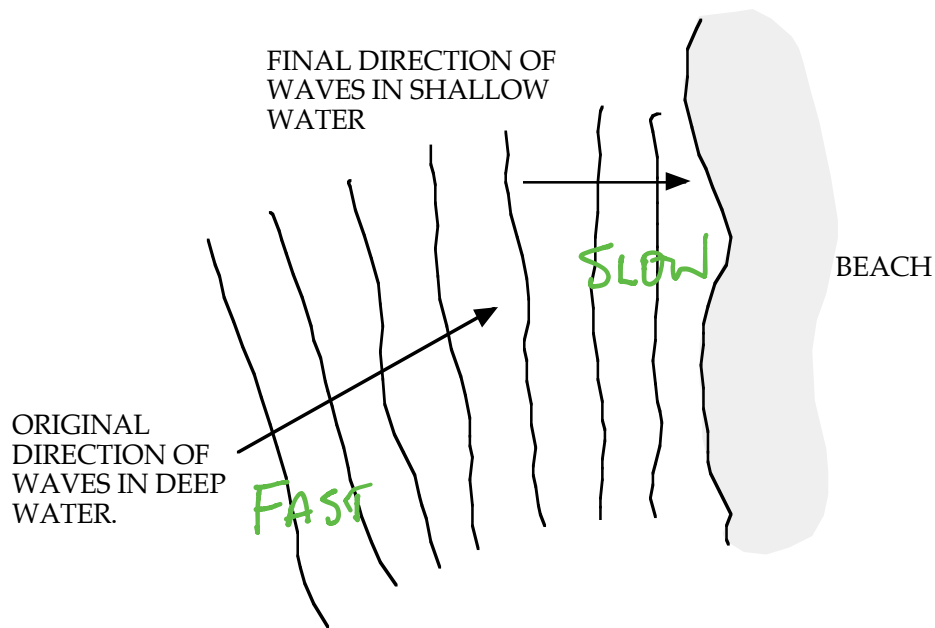


Figure 5.18

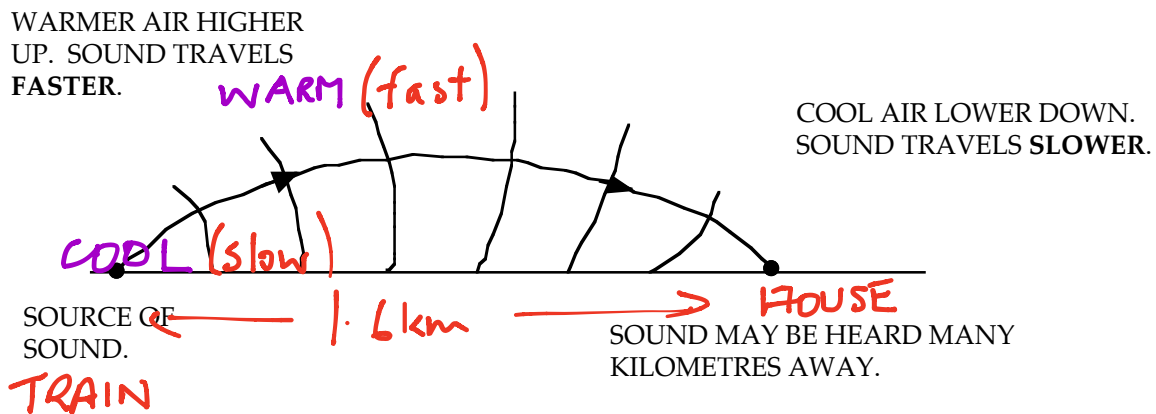
Sound waves being refracted at a boundary. i is the angle of incidence and r is the angle of refraction. The wavelength of the waves in the first material is λ_1 and in the second material it is λ_2 . The velocity of the waves in the first material is v_1 and in the second material is v_2 . The frequency of the sound is the same in both materials. (a) If sound waves from one medium enter another where the speed of sound is faster, they will be refracted away from the normal to the surface and their wavelength will be increased. (b) If sound waves from one medium enter another where the speed of sound is slower, they will be refracted towards the normal to the surface and their wavelength will be reduced.

Examples

1. Waves in the ocean slow down in shallow water and refract towards a normal to the beach, meaning that the waves arrive nearly parallel to the beach.



2. Sound is refracted by layers of air of differing densities, though the boundaries are ill-defined. When cooler air is trapped near the ground at night, sound can be reflected back down to the ground.



DIFFRACTION

Diffraction may be defined as the spreading out and change in direction of part of a wave as it passes around an obstacle or through an aperture in an obstacle.

It is at a **maximum** when the obstacle or aperture is about the **same size as the wavelength** of the incident wave.

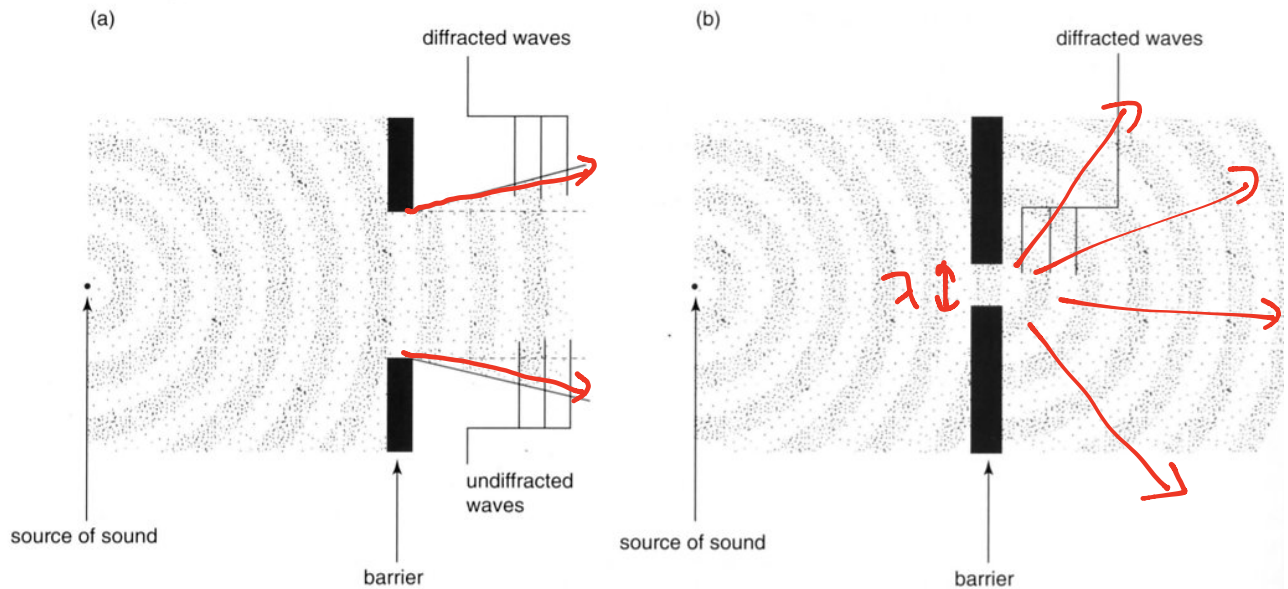
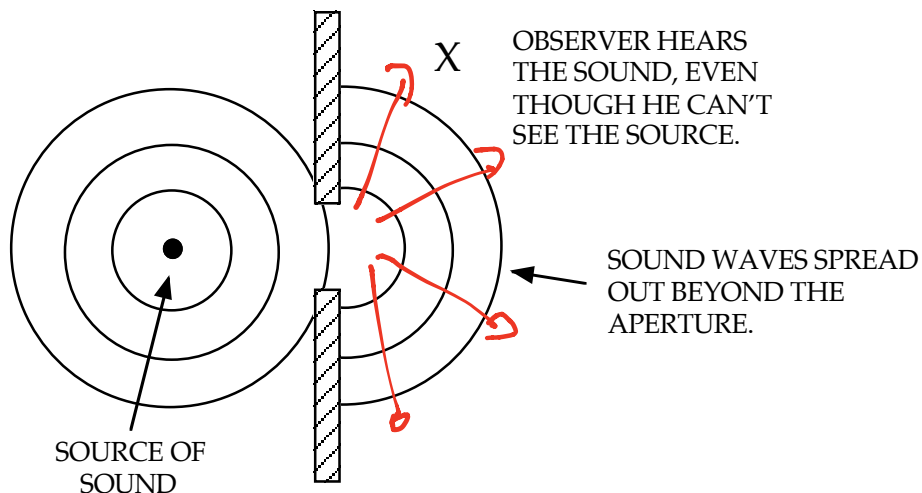


Figure 5.20

The amount of diffraction depends on the size of the gap, or aperture, and the wavelength. (a) When the aperture is much larger than the wavelength, only limited diffraction will occur. (b) Wavelengths larger than the aperture width will result in significant diffraction.

Examples

1. Sounds made outside a door can quite clearly be heard inside a room. The wavelengths of the sound waves are about the same size as the 1.00 m width of a standard door so the waves diffract into the room.



Reflection of sound from the walls will also help the sound reach the observer.

2. Ships and lighthouses have very low pitch horns so that they are heard clearly. Low sounds have wavelengths of several metres which is typical of distances between objects in a harbour.
3. Visible light waves have wavelengths of 4000 - 7000 Å (4.0×10^{-7} - 7.0×10^{-7} m). This distance is achieved on a diffraction grating, consisting of 600 - 1000 lines per millimetre ruled on a piece of glass or plastic. The distance between the lines corresponds to the wavelength of the incident light and diffraction occurs.
4. The wavelength of common sounds means that such sounds can bend around the corner of common objects that are quite large.

e.g. Sounds can be heard around the corners of buildings.

5.

When sound waves of a high frequency, for example 9000 Hz, strike an obstacle such as a person's head, they leave a distinct sound shadow in which little of the sound can be heard. If one ear is closer to the sound source than the other, these higher frequencies will be heard as louder by the ear closer to the source.

- a Assuming that the speed of the sound is 340 m s^{-1} , calculate the wavelength of a 9000 Hz sound.
- b Explain why this high-frequency sound leaves a sound shadow on one side of a person's head.

Solution

- a Using the wave equation:

$$v = 340 \text{ m s}^{-1}$$

$$f = 9000 \text{ Hz}$$

$$v = f\lambda$$

$$\lambda = \frac{v}{f} = \frac{340}{9000}$$

$$= 3.78 \times 10^{-2} \text{ m}$$

Hence, the wavelength is 0.0378 m or 3.78 cm.

- b A person's head is about 20 cm in diameter. The wavelength of a 9000 Hz sound (3.78 cm) is significantly smaller than this, and since diffraction is only significant when $\frac{\lambda}{w} \approx 1$ or more, diffraction will be minimal and the sound will not bend significantly around the head.

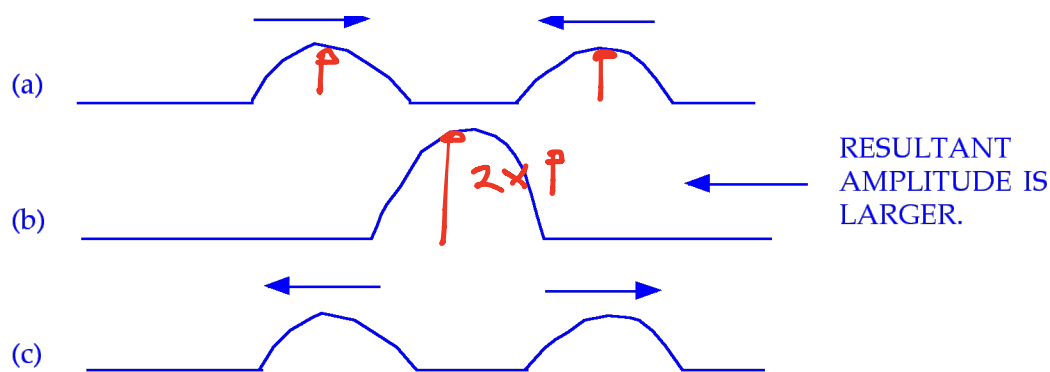
INTERACTIONS OF WAVES

Principle Of Superposition

When two or more waves travelling through the same medium at the same time meet, their displacements combine.

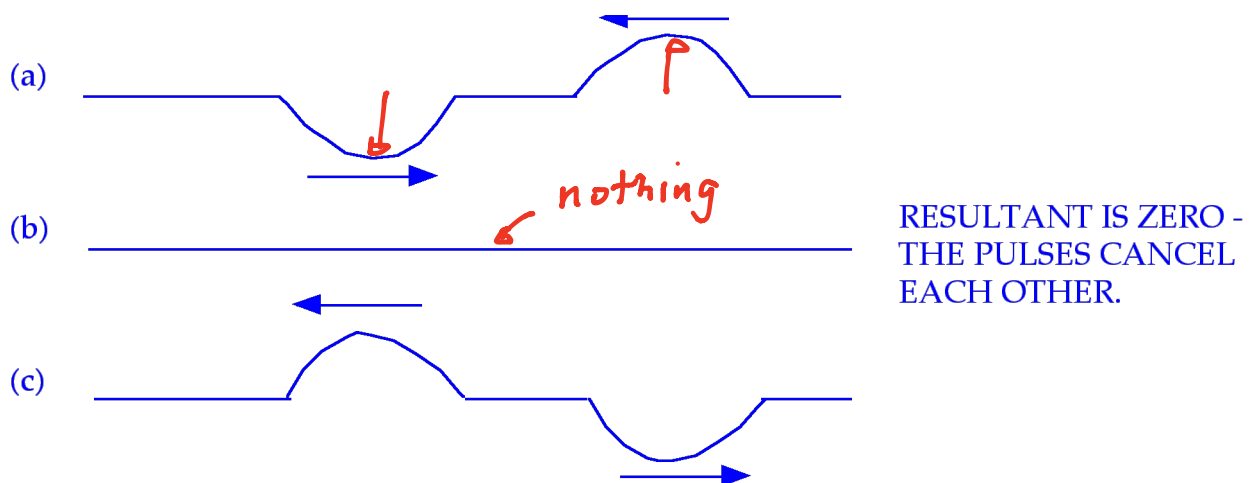
Waves can interfere *constructively* or *destructively*.

Constructive Interference: two pulses combine to give a pulse of greater amplitude.



- (a) Two pulses that are in phase to one another move in opposite directions in the same medium (a string).
- (b) When they meet, the resultant is the sum of both amplitudes.
- (c) The pulses continue unaffected.

Destructive Interference: two pulses combine to produce a pulse with a smaller amplitude than either of the two original amplitudes.



- (a) Two pulses 180° out of phase to each other are moving in the same medium.
- (b) Both pulses cancel to give a zero resultant.
- (c) The pulses continue unaffected.

Stationary Waves

These are produced when two identical waves of *equal amplitude, wavelength and speed* travel in *opposite directions* in the same medium.

The resulting wave has points of zero displacement (called **nodes**) and maximum displacement (called **antinodes**) occurring in the same position.

i.e. The wave is not “travelling” but “stationary”.

Example

Plucking a string on a guitar (or any string instrument) will send a pulse down it to a fixed end. Here it reflects 180° out of phase and moves back down the string to the opposite end (which is also fixed).

A stationary (or standing wave) is produced with its nodes and antinodes appearing not to move.

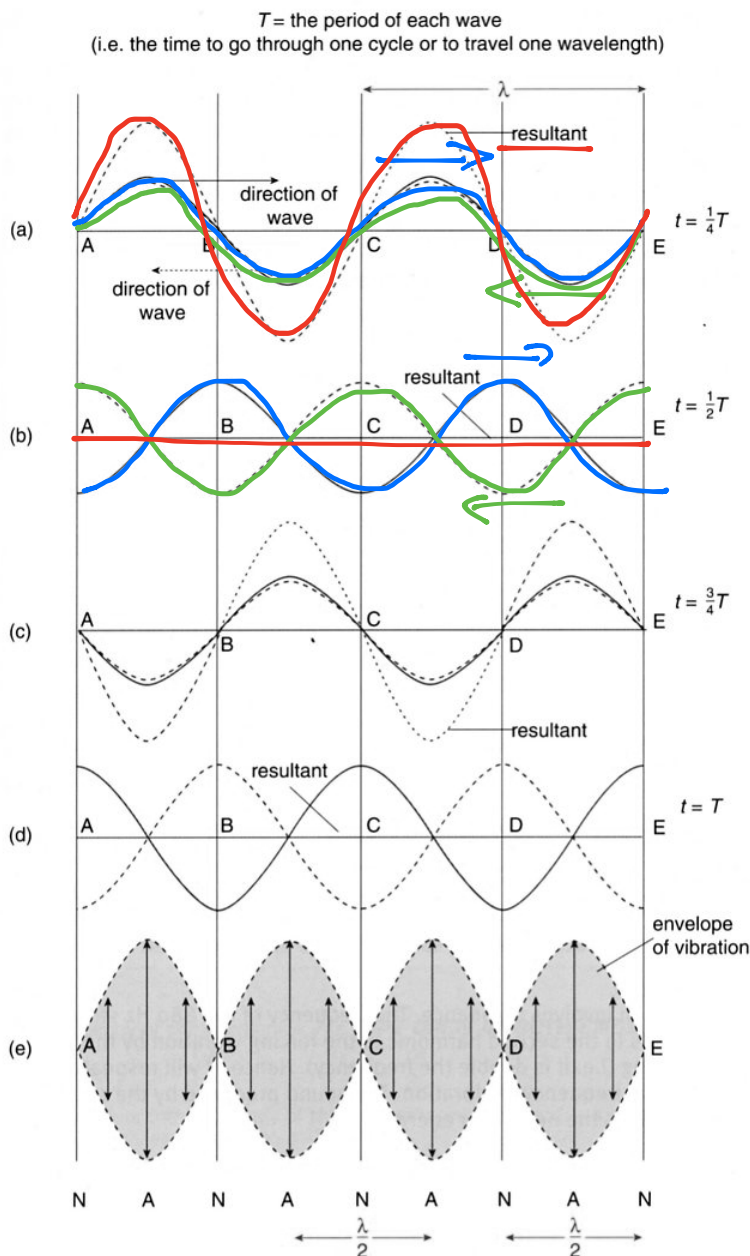


Figure 5.32

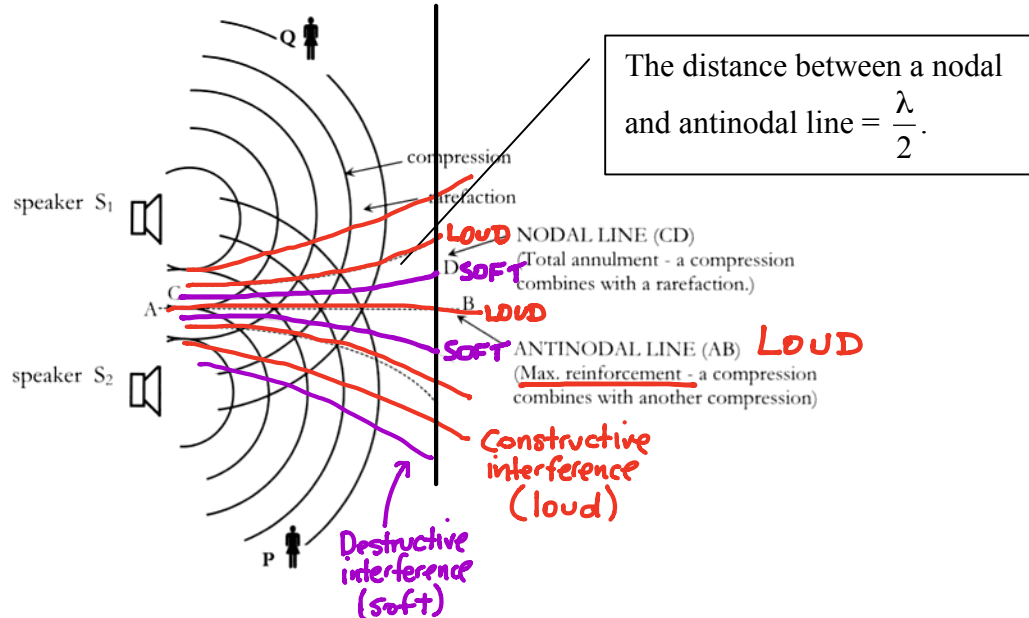
A standing (stationary) wave created in a string from two waves travelling in opposite directions, each with the same amplitude and frequency.

- (a) At a particular point in time, the two waves are completely superimposed so that crest meets crest and trough meets trough, to give the wave twice the original amplitude.
- (b) After a time equal to $T/4$ (one-quarter of the period) the waves will have moved $\lambda/4$, which means that they have moved $\lambda/2$ in relation to each other. The waves are completely out of phase and the resulting displacement is zero.
- (c) and (d) As further time goes by, the waves will continue to move past each other, creating the superimposed waveforms.
- (e) In the final wave envelope, the standing wave swings between maximum displacements, creating antinodes (A) which lie halfway between the stationary nodes (N). Regardless of the position of the component waves, these nodes stay in the same place. Successive nodal points lie $\lambda/2$ apart, as do successive antinodal points.

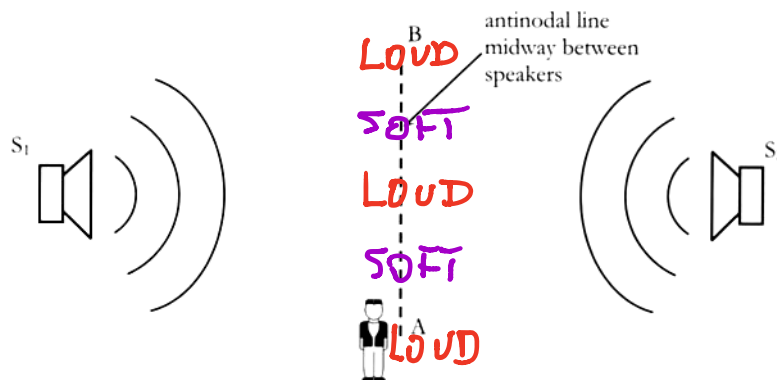
Interference From Two Point Sources

Consider two loudspeakers emitting the same sound (in phase) from two different points.

Points of maximum reinforcement (**antinodes**) and total annulment (**nodes**) will occur as shown in the diagram below.



A similar interference effect (nodes and antinodes) will occur when two loudspeakers are facing each other and emitting the same sound in phase. A standing wave pattern forms between the speakers.



- **Antinodal lines** occur where there is **constructive** interference - a **loud sound** is heard.
- **Nodal lines** occur where there is **destructive** interference - a **soft sound** is heard.

The difference in distance from either speaker is called the **path difference**. There are two relationships to describe when constructive and destructive interference occur.

Constructive Interference: $\text{path difference} = n\lambda$

where $n = 0, 1, 2, 3, \dots$

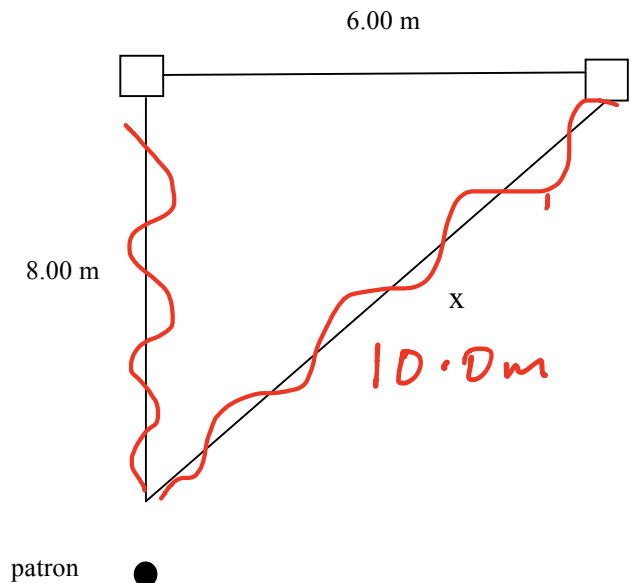
Destructive Interference: $\text{path difference} = \frac{n\lambda}{2}$

where $n = 1, 2, 3, \dots$

*180° out of phase
= $\frac{\lambda}{2} \times n$*

Examples

- In a cinema, a patron is sitting in front of two large speakers as shown. Assume the speakers are in phase with each other.



- Calculate the path difference of the sound heard by the patron.

$$x = \sqrt{(6.0)^2 + (8.0)^2}$$

$$= 10.0 \text{ m}$$

$$\therefore \text{path difference} = 10.0 - 8.0$$

$$= \underline{2.0 \text{ m}}$$

- If the speakers are both producing 680 Hz, what does the patron hear?

$$v = f\lambda$$

$$\Rightarrow \lambda = \frac{v}{f}$$

$$= \frac{340}{680}$$

$$= \underline{0.50 \text{ m}} \quad \checkmark$$

$$\text{path difference} = \underline{2.0 \text{ m}}$$

$$= 4\lambda$$

$$\Rightarrow \text{constructive interference}$$

$$\therefore \underline{\text{Patron hears a loud sound.}}$$

$\frac{2.0}{0.5} = 4 \times \lambda$

*Calculate λ ,
compare to path difference*

- (c) What is the lowest frequency needed for a soft sound? soft sound
 $\Rightarrow \frac{\lambda}{2}$ difference

Lowest frequency $\Rightarrow$ Large λ

$$\begin{aligned}\text{Minimum path difference} &= \frac{\lambda}{2} = 2.0 \text{ m} \\ \Rightarrow \lambda &= 4.0 \text{ m}\end{aligned}$$

$$v = f\lambda$$

$$\begin{aligned}\Rightarrow f &= \frac{v}{\lambda} \\ &= \frac{340}{4.0} \\ &= \underline{85.0 \text{ Hz}}\end{aligned}$$

2.

Two loudspeakers (A and B) are set up as shown in Figure 1.14(a) so that they both emit sounds of 425 Hz which are in phase. The speakers are 2.40 m apart. Assume the speed of sound is 340 ms⁻¹.

- What is the wavelength of the sounds emitted?
- Melissa stands exactly midway between the speakers and then walks away so as to always remain equidistant from them. Describe what she hears. Explain why.
- Melissa now walks to a point C, 7.0 m directly in front of speaker A.
 - How far is she from speaker B?
 - What will be the effect?

$$v = 340 \text{ ms}^{-1}$$

$$f = 425 \text{ Hz}$$

$$\lambda = ?$$

$$AB = 2.40 \text{ m}$$

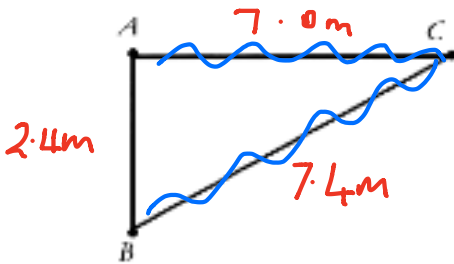
$$(a) \quad v = \lambda f \quad \therefore \lambda = \frac{v}{f} = \frac{340}{425}$$

$$\lambda = 0.80 \text{ m}$$

- (b) If Melissa is always equidistant from the speakers the sounds reaching her will always be in phase. She will therefore always hear a loud sound.

$$\begin{aligned}(c) \quad AB &= 2.40 \text{ m} & (BC)^2 &= (AC)^2 + (AB)^2 \\ AC &= 7.00 \text{ m} & &= 49 + 5.76 \\ BC &= ? & \therefore BC &= 7.40 \text{ m}\end{aligned}$$

Since the wavelength of the sound is 0.80 m then at point C the sounds are exactly half a wavelength apart (180° out of phase). Melissa will "hear" silence.



$$p.d. = 7.4 - 7.0$$

$$= 0.4 \text{ m}$$

$$= \frac{\lambda}{2}$$

$\therefore$ soft sound

PRODUCTION OF SOUND

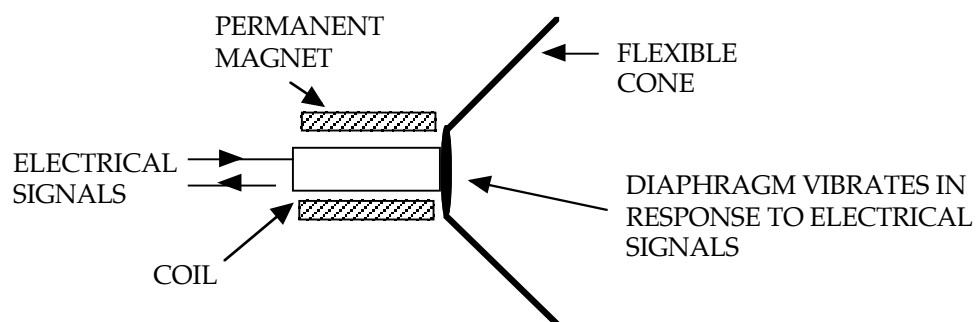
A sound wave can be generated by a vibrating object imparting a longitudinal wave motion to the surrounding medium.

The particles of the medium receive energy from the vibrating source and begin to vibrate at the same frequency.

The two fundamental requirements for the production and propagation of sound are a vibrating source and an elastic medium.

Example

A loudspeaker.



The coil moves in response to electrical signals as per the motor principle - its magnetic field interacts with the permanent magnet's field.

The diaphragm is connected to the coil and moves accordingly.

Sound is produced by the diaphragm vibrating, causing the cone to push against the air particles and transfer energy to them.

As the cone moves forwards, a compression is produced. As it moves backwards, a rarefaction is produced.

SPEED OF SOUND

The velocity of sound in a medium is determined by the **elasticity** and **density** of a medium.

The greater the elasticity, the greater the velocity.

The greater the density, the slower the velocity.

The velocity of sound in air is affected by temperature but not pressure.

i.e. Increasing temperature increases the velocity.

$$v_{\text{sound}} = 331 \text{ m s}^{-1} \text{ (at } 0^\circ\text{C)} \pm 0.6 \text{ m s}^{-1} \text{ for every } 1^\circ\text{C change.}$$

FREQUENCY AND PITCH

Increasing the frequency increases the pitch of the sound.

The pitch of a sound is the *perception* a listener has of a sound - whether it is high or low.

The human ear has a frequency range of 20 - 20,000 Hz.

Frequencies above 20,000 Hz are called *ultrasonic*. Dogs hear up to 50,000 Hz while bats respond up to 100,000 Hz.

Those less than 20 Hz are called *infrasonic*. These frequencies are generated by earthquakes and heavy machinery vibrating.

CHARACTERISTICS OF SOUND WAVES

Sounds can be classified as either musical notes or noise.

Musical Notes: vibrating body produces a regular frequency and a definite pitch.

Noise: produced by a body that does not vibrate with a regular frequency.

Stretched strings and columns of air vibrate in particular ways to produce sound.

Vibrations In Stretched Strings

As seen before, a standing wave is set up in a string that is attached at both its ends.

A string will vibrate in different ways according to how it is plucked, strummed, struck or where a bow is pulled across it.

i.e. In the middle, down towards one end, etc.

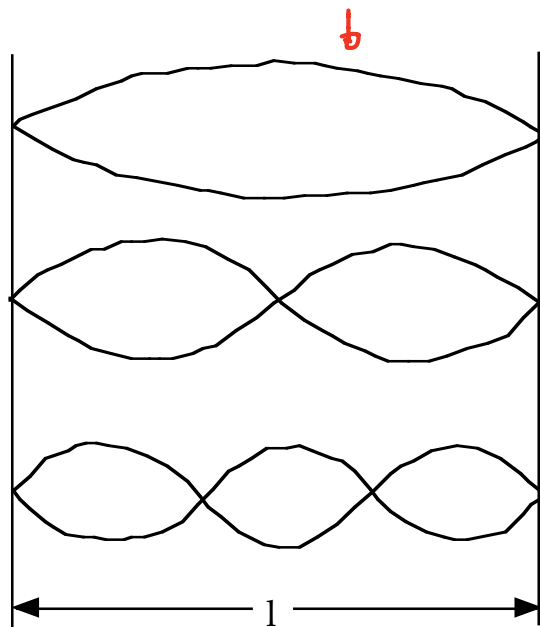
The lowest frequency at which a string will vibrate is called the *fundamental frequency* (or *1st harmonic*).

The next mode of vibration is called the *second harmonic* and has a *frequency that is twice that of the fundamental frequency*.

Subsequent harmonics have frequencies that are whole number multiples of the fundamental frequency.

Harmonic: A whole number multiple of the fundamental frequency.

Learn to draw this!



HARMONIC	1	f
1st	$\frac{\lambda}{2}$	f_1

2nd	$\frac{2\lambda}{2}$	$2f_1$
-----	----------------------	--------

3rd	$\frac{3\lambda}{2}$	$3f_1$
-----	----------------------	--------

$$l = \frac{3}{2} \lambda \Rightarrow \lambda = \frac{2l}{3}$$

NOTICE THAT THE HARMONICS HAVE WHOLE-NUMBERED MULTIPLES OF THE FUNDAMENTAL LENGTH (l) AND FREQUENCY (f).

In strings, **all** of the harmonics are possible.

i.e. $n = 1, 2, 3, 4, \dots$

The frequency of the harmonics in a string is given by:

Remember!

$$f = \frac{nv}{2l}$$

where
 f = frequency of vibration (Hz).
 n = number of the harmonic.
 v = speed of sound (ms^{-1}).
 l = length of the string (m).

$$\lambda = \frac{2l}{n} \Rightarrow v = f\lambda \Rightarrow v = \frac{f2l}{n} \Rightarrow f = \frac{nv}{2l}$$

Examples

1.

When the key on a piano corresponding to a note of 440 Hz is struck firmly, it is found that the 880 Hz string also vibrates, even though the two are not connected and are some distance apart. How can this be so?

Solution

This situation involves resonance. The frequency of the 880 Hz string corresponds to the second harmonic of the forcing vibration by the 440 Hz string (i.e. it is double the frequency). Hence, it will resonate at its natural frequency of vibration. The sound produced by the 440 Hz string provides the necessary energy.

2.

A violin string has a length of 22.0 cm. It is vibrating with its fundamental mode of vibration at a frequency of 880 Hz. What is the wavelength of this fundamental frequency?

Solution

The fundamental frequency corresponds to the first harmonic. That is:

$$n = 1$$

$$\lambda_n = \frac{2L}{n}$$

$$L = 22.0 \times 10^{-2} \text{ m}$$

$$\lambda_1 = \frac{2(22.0 \times 10^{-2})}{1} = 4.40 \times 10^{-1} \text{ m}$$

Air Columns

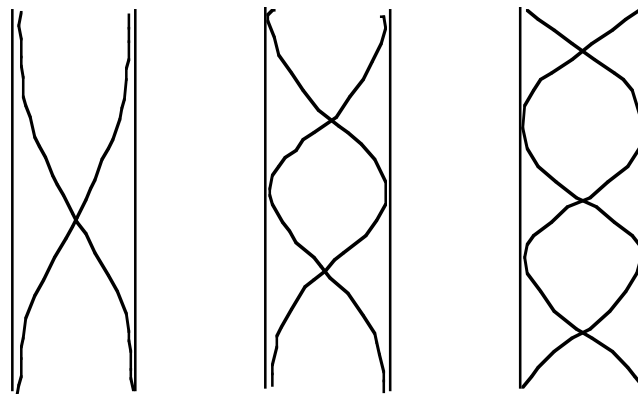
The air in an open or closed pipe will resonate to a vibrating source placed at one end.

e.g. A tuning fork, blowing across the top of the pipe (the turbulence causes the vibrations), a vibrating reed in a clarinet.

A longitudinal standing wave is set up in the pipe with reflection occurring from each end, regardless of whether it is the open or closed end.

At an *open end*, only an *antinode* is possible, and at a *closed end* only a *node* is possible.

Open Pipes



HARMONIC 1st

2nd

3rd

$$l$$

$$1/2 \lambda$$

$$\approx \frac{2\lambda}{2}$$

$$3/2 \lambda$$

$$f$$

$$f_1$$

$$2f_1$$

$$3f_1$$

For open pipes, *all harmonics are possible*.

The fundamental length of the pipe is $l = 1/2 \lambda$.

The distance between successive harmonic lengths is $1/2 \lambda$.

The frequency of the harmonics in an open pipe is given by:

$$f = \frac{nv}{2l}$$

Applies to open pipes + strings

where f = frequency of vibration (Hz).
 n = number of the harmonic.
 v = speed of sound (ms^{-1}).
 l = length of the open pipe (m).

Examples

1. An open pipe has an effective length of 1.23 m. If the speed of sound in air is $3.40 \times 10^2 \text{ ms}^{-1}$, calculate:

- (a) the fundamental frequency. *1st harmonic*
 (b) the frequency and wavelength of the third harmonic.

(a) For the first harmonic:

$$l = \frac{1}{2} \lambda$$

$$\Rightarrow \lambda = 2(1.23)$$

$$= 2.46 \text{ m}$$

$$c = f\lambda$$

$$\Rightarrow f = \frac{c}{\lambda}$$

$$= \frac{3.40 \times 10^2}{2.46}$$

$$= 1.382 \times 10^2 \text{ Hz}$$

$\therefore$ Fundamental frequency = $1.38 \times 10^2 \text{ Hz}$.

(b) For the third harmonic:

$$l = \frac{3}{2} \lambda$$

$$\Rightarrow \lambda = \frac{2}{3}(1.23)$$

$$= 0.820 \text{ m}$$

$$f_3 = 3f_1$$

$$= 3(1.38 \times 10^2)$$

$$= 4.14 \times 10^2 \text{ Hz}$$

$$f = \frac{v}{\lambda}$$

$$= \frac{340}{0.820}$$

$$= 414 \text{ Hz}$$

$\therefore$ Wavelength = 0.820 m and frequency = $4.14 \times 10^2 \text{ Hz}$.

2.

A particular flute has an effective length of 35.0 cm. It can be thought of as an open-ended air column. The sound in the tube has a velocity of 350 m s^{-1} .

- a** What is the wavelength of the second harmonic?
b What will the frequency of the second harmonic be?

Solution

- a** For the second harmonic, a drawing of the appropriate standing wave will show that:

$$L = 35.0 \times 10^{-2} \text{ m}$$

$$n = 2$$

$$\lambda_n = \frac{2L}{n}$$

$$\lambda_2 = \frac{2(35.0 \times 10^{-2})}{2}$$

$$= 3.50 \times 10^{-1} \text{ m}$$

- b** Using the wave equation and rearranging:

$$n = 2$$

$$f_n = \frac{nv}{2L}$$

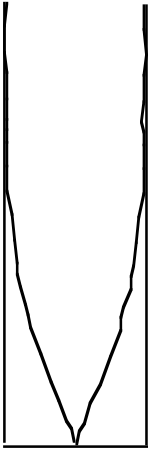
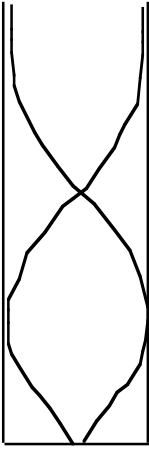
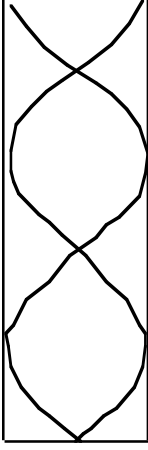
$$L = 35.0 \times 10^{-2} \text{ m}$$

$$f_2 = \frac{2(350)}{2(35.0 \times 10^{-2})}$$

$$v = 350 \text{ m s}^{-1}$$

$$= 1.00 \times 10^3 \text{ Hz}$$

Closed Pipes

			
HARMONIC	1st	3rd	5th
l	$1/4 \lambda$	$3/4 \lambda$	$5/4 \lambda$
v	v_1	$3v_1$	$5v_1$

For closed pipes, **only odd harmonics occur.**

The fundamental length is $l = 1/4 \lambda$.

The distance between successive harmonic lengths is $1/2 \lambda$.

The frequency of the harmonics in a closed pipe is given by:

$$f = \frac{nv}{4l}$$

← Note the 4!

Examples

1.

The ear canal, from the outer ear to the eardrum, can be thought of as a tube closed at one end (by the eardrum) and open at the other. It is approximately 3.00 cm long in an adult. Assume that the speed of sound is 340 m s^{-1} .

- a What is the fundamental resonant frequency of the ear canal?
- b What is the frequency of the next resonating frequency?
- c Explain one reason why some frequencies are heard better than others.

Solution

- a It is useful to refer to a diagram of the situation for a particular harmonic in order to determine the relationship between λ and L . Figure 5.36 shows that for a closed tube, the length will correspond to $L = \frac{\lambda}{4}$ and so:

$$v = 340 \text{ m s}^{-1}$$

$$f_n = \frac{nv}{4L}$$

$$L = 3.00 \times 10^{-2} \text{ m}$$

$$f_1 = \frac{1(340)}{4(3.00 \times 10^{-2})}$$

$$n = 1$$

$$= 2.83 \times 10^3 \text{ Hz}$$

Compare this with the frequencies heard best by humans.

- b As this is a closed tube, only odd-numbered harmonics will form, so the next harmonic will be the third:

$$v = 340 \text{ m s}^{-1}$$

$$f_n = \frac{nv}{4L}$$

$$L = 3.00 \times 10^{-2} \text{ m}$$

$$f_3 = \frac{3(340)}{4(3.00 \times 10^{-2})}$$

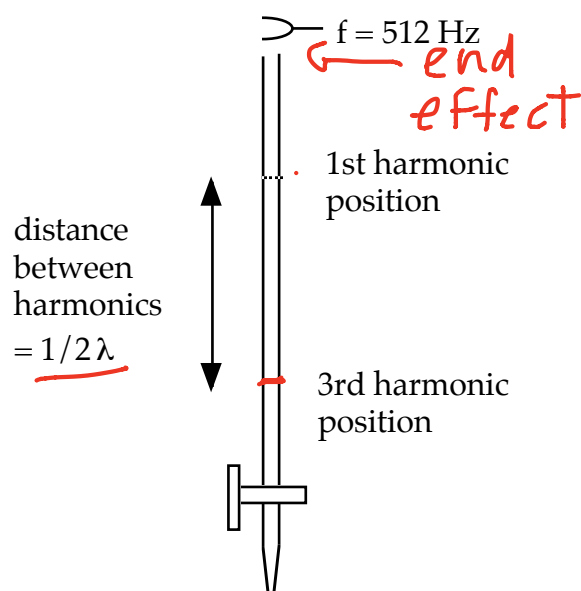
$$n = 3$$

$$= 8.50 \times 10^3 \text{ Hz}$$

- c The standing waves of resonant frequencies will have substantially larger amplitudes than the sounds that do not cause resonance. As a result, sounds which closely correspond to resonant frequencies, like those calculated above, will be amplified more and have more chance of being heard than others. Of course there are many other factors which influence the range of frequencies a particular person is able to hear.

2. A group of students performed an experiment in the laboratory to determine the velocity of sound in air. They filled a burette with water and sounded a 512 Hz tuning fork over the top of it as they let the water slowly flow out the bottom. They found two successive resonant positions when the air column was 0.170 m and 0.505 m from the top.

Using this information, calculate the value for the speed of sound the students obtained.



The first position is not used for calculating the speed of sound since there is an “end effect” that has to be taken into account. This is due to the particles near the open end being able to move in all directions rather than one dimension.

The distance between the two resonant positions is equal to $1/2 \lambda$.

$$\begin{aligned} \text{i.e. } 0.335 &= 1/2 \lambda \\ \Rightarrow \lambda &= 2(0.335) \\ &= 0.670 \text{ m} \end{aligned}$$

$$\begin{aligned} c &= f \lambda \\ &= (512)(0.670) \\ &= 3.430 \times 10^2 \text{ ms}^{-1} \end{aligned}$$

$\therefore$ The speed of sound = $3.43 \times 10^2 \text{ ms}^{-1}$.

MUSICAL SOUNDS

Musical sounds have three distinct characteristics - *pitch, loudness and quality*.

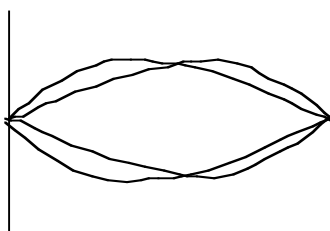
When a piano and a flute play the same note having the same pitch and loudness, there is a clear difference between the sound produced by each instrument. There is a *difference in the quality* of the sounds.

The quality of a musical note depends on the *number, nature and relative intensities of the harmonics* present in the note.

The greater the number of harmonics present in a note, the better its quality.

Harmonics depend on the mode of vibration of a stretched string or an air column.

e.g. Consider a stretched string.



A STRETCHED STRING VIBRATING WITH ITS FUNDAMENTAL AND SECOND HARMONIC SIMULTANEOUSLY.

Physics in action — Music or noise?

What is music to one person may be just plain noise to another. The frequencies that are present in sounds vary, and any clear distinction between music and noise is mainly a subjective judgement. Physics can provide a basic distinction between what we define as a harmonious sound and as noise, but it does not attempt to draw a line between the two.

Related and unrelated frequencies

Musical instruments generally produce sounds which are whole-number multiples of a fundamental frequency. When a sound from a musical instrument is displayed on an oscilloscope or a digitiser, a clear and distinctive stable wave pattern is seen. This is because the vibrations produced by most musical instruments are only free to vibrate in one dimension. A sound which includes these simple multiples of the fundamental frequency is usually regarded as harmonious.

Hitting two sticks together creates a different mixture of frequencies. The vibrations within the sticks that cause the sound move in three dimensions, and the range of frequencies produced is more complex. The display on a digitiser or an oscilloscope would not show a stable wave pattern; the frequencies bear little or no relation to each other. The sound may still be quite pleasant to hear, but it can be interpreted as noise since it consists of a random mixture of frequencies. 'White noise' is

such a mixture of frequencies. Although we define it as noise, it can be soothing and is quite a pleasant sound to many people.

Instruments such as drums lie between the extremes of simple multiples of frequencies and random mixtures. Drums can vibrate in two dimensions, so they do not have the pure harmonious sound of other instruments, but the frequencies do bear some relation to each other. Totally enclosed drums can still sound harmonious, because certain frequencies resonate and are accentuated.

Harmonies and musical scales

Harmonies are an important part of music. Two notes generally sound harmonious if the ratio of their frequencies is a simple whole number. The simplest ratio of 2:1 sounds the most harmonious: this is what we call an octave. A ratio of 3:2, called a 'fifth' since it includes five musical intervals (tones and semitones), also sounds very harmonious. Combinations based on more complex ratios, such as 5:4, sound 'dissonant'. Modern music has seen the acceptance of more dissonant combinations, and today almost any dissonance can be found in a musical score. Many different musical scales involving these ratios have been used throughout the world. For example, African music uses much smaller intervals between notes and thus has two to four times as many defined notes as traditional Western music.





Figure 5.37

Because the vibration of a drum skin occurs in two dimensions, in contrast to the single longitudinal dimension of a string, it creates less harmonious sounds than many other musical instruments.

The most common modern scale is called the 'equally tempered chromatic scale'. This evolved from a scale based on four intervals corresponding to simple fractions of the length of a vibrating string. Various versions of the earliest chromatic scale led to a compromise that suits keyboard instruments such as the piano. (Earlier versions had many more intervals within one octave and would have required a large and unplayable keyboard.) The equally tempered chromatic scale divides a simple 2:1 ratio, one octave, into 12 equal semitones. Each pair of adjacent notes thus has the same frequency ratio.

As an octave is a basic ratio of 2:1, having 12 equal intervals means that the ratio of the frequency of any one note on a piano to that of the next will be $\sqrt[12]{2:1}$ or about 1.06:1.

table 5.1 Frequencies of common notes on the equally tempered chromatic scale, based on A = 440 Hz. The tonal names refer to the seven white notes on the keyboard. Compromises in the scale mean that adjacent sharps (#) and flats (b) have the same frequency. (Upper C actually has exactly double the frequency of middle C. It is noted here as 523 and not 524 Hz because the values are rounded to the nearest whole number.)

Note	Tonal name	Frequency (Hz)
A		220
A [#] , B ^b		233
B		247
C (middle)	do	262
C [#] , D ^b		277
D	re	294
D [#] , E ^b		311
E	mi	330
F	fa	349
F [#] , G ^b		370
G	so	392
G [#] , A ^b		415
A	la	440
A [#] , B ^b		466
B	ti	494
C	do	523

FORCED VIBRATIONS

Objects such as a steel rod, tuning fork or a glass can be made to vibrate by striking them.
i.e. A single force is applied for a short time.

These objects will vibrate *at their natural frequency* (which depends on their physical dimensions and the material of which they are made).

However, if they are made to vibrate by an application of a periodic driving force (i.e. one applied with a regular frequency) with the same frequency as that of the driving force, this is called *forced vibration*.

Example

A tuning fork that is struck and held in the hand is very difficult to hear unless it is placed close to the ear.

If it is struck and then placed in contact with a table, it is heard very clearly.

The table is forced to vibrate at the same frequency as the tuning fork. A louder sound is produced because a much larger mass of air in contact with the table is made to vibrate for a shorter period of time.

In musical instruments such as violins and guitars, the wooden frame and the enclosed air columns are forced to vibrate with the same frequency as the strings, thereby increasing the sound intensity.

RESONANCE

When the frequency of the periodic driving force coincides with the natural frequency of the body being forced to vibrate, a large amplitude of vibration is developed.

This phenomenon is called *resonance*.

Examples

1. A child's swing. Small pushes given when the swing is about to move back from its maximum displacement will greatly increase the amplitude of the swing.
2. Stretched strings or air columns have a natural frequency of vibration. Sound waves of the same frequency produced by a vibrating source will cause them to resonate.
3. Parts on a car will vibrate or rattle at particular speeds. The vibrations from the tyres on the road match the natural frequency of the parts and cause them to vibrate significantly.
4. Dipping a finger in wine and rubbing it gently around the top of a wine glass will cause it to "ring". The ridges of the fingerprint cause small vibrations that match exactly the natural frequency of the air column inside the glass so it resonates.

BEATS

Beats refer to the unusual waxing and waning of sound that is produced when two sounds of similar amplitude but slightly differing frequencies occur at the same time.

i.e. *The sound rises and falls in a regular pattern.*

Beats are another example of interference between sound waves. The two waves superimpose and produce a single wave whose amplitude varies with time.

The resultant single wave is generated by the algebraic sum of the displacements of the individual waves.

Physics in action — Some effects of superposition

Superposition creates problems for synthesised sound, but it can also be used to our advantage.

Beats

If two sound waves of equal amplitude but slightly different frequency are combined, the resulting sound has a regular pulsation, called a *beat*. The effect is only noticeable when the two frequencies are close to each other. This phenomenon is the direct result of superposition of the sound waves.

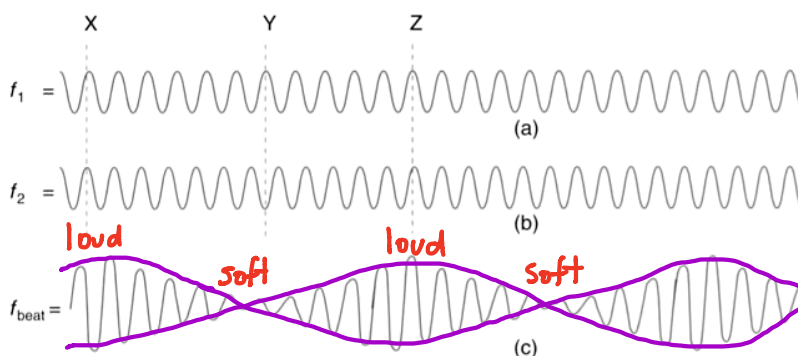


Figure 5.25

A beat (f_{beat}) produced by two sound waves f_1 and f_2 of equal amplitude but slightly different frequency. At time X the waves are in phase, producing a larger amplitude. A short time later, at Y, they are out of phase, resulting in a smaller amplitude, and at Z they are in phase once again. When the two waves are in phase (at X and Z), the resulting pressure variation, and hence volume, is large. At Y the waves are out of phase and the resulting amplitude and volume are zero.

Beats can be explained by looking at the graphs of pressure variation with time for two waves of the same amplitude and a slightly different frequency over time (Figure 5.25).

The frequency of the beats (the *beat frequency*) is simply the difference between the frequencies of the sounds that are superimposed to cause the beats:

$$f_{\text{beat}} = |f_1 - f_2|$$

Beats are a reliable means of detecting small differences in the frequency of two sounds, and are therefore useful in tuning musical instruments. While the exact frequency of a note may be hard to determine, notes that are not exactly the same will produce a beat. The instruments are tuned so that the beat gradually slows down, indicating that the two instruments are getting closer to the same frequency. When one sound of constant volume is heard, the instruments are at the same frequency. (Of course, this method does not guarantee that the frequency will be the right one unless a standard tone, electrical tuner or tuning fork is used as a control.)

The frequency at which beats are produced is given by:

$$f_{\text{beat}} = |f_2 - f_1|$$

Use logic to remove " | | "

where f_{beat} = the beat frequency of the resultant wave (Hz)
 f_1, f_2 = the frequency of the two waves generated at the same time (Hz)

Example

U slower
 $f = 360 \text{ Hz}$
366.5 Hz ?
353.5 Hz ?

When a piano note and a tuning fork of frequency 360 Hz are sounded together, 6.5 beats per second are heard. One prong of the tuning fork is loaded with a small piece of wax and the number of beats heard per second is increased. What is the frequency of the piano note?

Putting wax on the fork will slow down the tines so its frequency is lowered.

Since this increases the beat frequency, then the piano note must be higher than the fork.

$$\begin{aligned}\therefore f_{\text{beat}} &= f_{\text{piano}} - f_{\text{fork}} \\ \Rightarrow f_{\text{piano}} &= 6.5 + 360 \\ &= 366.5 \text{ Hz}\end{aligned}$$

$\therefore$ The piano note is 366.5 Hz.

tuning fork } 6.5 Hz
piano }
tuning fork } 6.5 Hz ↑ ✓✓
↓